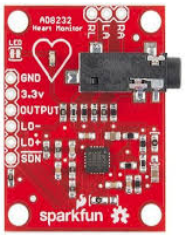
AI - DRIVEN PLANT ELECTROPHYSIOLOGY FOR INTELLIGENCE CROP MONITORING

1. Project Definition

This project introduces an **AI-Based Crop Silent Stress Detection System** that utilizes plant bio-signals to identify early stress conditions. Unlike traditional methods that rely on visible symptoms like wilting or discoloration, this system monitors internal "plant electrophysiology" using micro-voltage sensors [cite: 8, 44, 50]. By analyzing these bio-potential signals alongside environmental data, the system predicts stress types such as drought or disease before they cause irreversible damage, enabling timely intervention and sustainable farming [cite: 52, 53, 72].

2. System Components

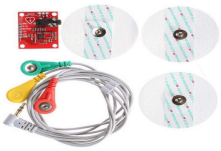


Instrumentation Amplifier (AD8232)

Captures the high-frequency plant bio-potential signals (μV) and filters out environmental electrical noise [cite: 4, 104].

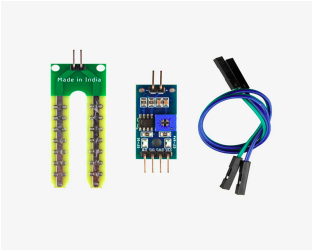
ESP32 Microcontroller

The central hub for data processing, signal normalization, and wireless connectivity to the cloud [cite: 5, 97, 104].



Silver/Silver-Chloride (Ag/AgCl) Electrodes

Provides a stable, non-invasive connection to the plant stems for precise electrical signal acquisition [cite: 3].



Environmental Sensors (Soil & DHT11/22)

Measures soil moisture, temperature, and humidity to cross-reference with plant bio-signals for accurate diagnosis [cite: 6, 7, 109].

3. Tech Stack

The following languages and frameworks form the core of the PhytoPulse platform:

Python | Embedded C | JavaScript (Vanilla) | HTML5 & CSS3 | Tailwind CSS | SQL (MySQL) | Flask (API)

Note: Used for microcontroller programming, ML model training (Scikit-learn), and responsive web visualization [cite: 88, 89, 90, 91].

4. Usage & How to Use

Step 1: Hardware Setup - Connect the electrodes to the plant stem using ECG gel and insert the moisture probe into the soil. Power the ESP32.

Step 2: Signal Initialization - Access the PhytoPulse dashboard. The system will initialize sensors and begin reading real-time bio-potential signals [cite: 104].

Step 3: Monitoring & Alerts - View the live waveform graph. The AI engine will automatically classify the plant's health status (Healthy / Stressed / Disease) [cite: 104].

Step 4: Data Export - Use the "One-Click Export" feature to download sensor history in CSV format for further research.

5. Future Enhancement

- **Automated Response:** Integration with IoT-based irrigation and electrostimulation therapy [cite: 104, 121].
- **Advanced Intelligence:** Implementing deep learning models and weather/satellite data integration [cite: 123, 124].
- **Mobility:** Launching a dedicated mobile application for real-time farmer alerts [cite: 122].